

Lab-8 Ex3 - GCD & LCM

Write a program to read 4 integers (you may assume they are all positive integers), and then output their Greatest Common Divisor (GCD) and their Least Common Multiple (LCM).

In this exercise, you are given a function `gcd()` that computes and returns the GCD of two integers. Your objective is to implement a function that computes and returns the LCM of two integers, and then use these functions to compute the GCD and LCM of the four input values.

Let `gcd(x, y)` be the GCD of `x` and `y`. Then the LCM `x` and `y` can be computed as

$$\text{lcm}(x, y) = (x * y) / \text{gcd}(x, y)$$

Let `a, b, c, d` be four positive integers. Then one way to compute the GCD and LCM of these four numbers are as follows:

- `gcd(a, b, c, d) = gcd( gcd( gcd(a, b), c), d)`
- `lcm(a, b, c, d) = lcm( lcm( lcm(a, b), c), d)`

You may verify your results using any online calculator.

User-input numbers are ***bold and italic*** in the following sample runs.

Sample Run #1

10 5 15 20

GCD is 5

LCM is 60

Sample Run #2

2 4 8 16

GCD is 2

LCM is 16

Sample Run #3

2 3 5 7

GCD is 1

LCM is 210